

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Pesticides - Set 1

Mark scheme

AQA spec 3.4.3.2.8; Pollution; textbook pp. 262-266

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• 25 g ha ⁻¹ (accept 24-27)	1
01.2	• The highest dose that can be applied without killing any members of the population / any individuals of the (non-target) species	1
01.3	<i>Any 3 from:</i> • The MDAF (minimum dose always fatal) of the pest is about 50 g ha⁻¹ • This is higher than the MDNF of the bee (about 20 g ha⁻¹) • So a dose that kills all pests kills some bees / the two curves overlap between about 20 and 50 g ha⁻¹ • A dose of 30-40 g ha⁻¹ would kill most pests with limited bee deaths (compromise)	3
01.4	<i>Any 2 from:</i> • Spray at night / when bees are not flying • Spray when the crop is not in flower • Spray on still days to reduce spray drift • Use a systemic pesticide / seed coating rather than spraying • Use smaller droplets for more even coverage	2
01.5	<i>Any 3 from:</i> • Specificity is the extent to which the pesticide is toxic only to the target species / range of species affected • A highly specific pesticide has low toxicity to non-target species (eg bees, fish, mammals) • So fewer non-target deaths / less loss of predators, pollinators • Fewer indirect (food chain) effects / fewer inter-species relationships lost • Example: pyrethroids have high insect toxicity but low mammal toxicity	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-263; AQA spec 3.4.3.2.8.	

Question	Marking guidance	Marks
02.1	<i>Any 4 from:</i> • High persistence means DDT remains in the environment for years / decades, so it has time to disperse and be taken up • High liposolubility means it dissolves in fat and is stored in body tissues rather than excreted (bioaccumulation) • Concentration increases at each trophic level as predators eat many contaminated prey (biomagnification) • Birds of prey are at the top of the food chain so receive the highest concentrations • Sub-lethal doses caused thin eggshells that broke / reduced breeding success; higher doses caused sterility or death	4
02.2	• Advantage: low persistence / low liposolubility so they do not bioaccumulate or biomagnify / break down quickly • Disadvantage: high mammalian toxicity so farm workers / humans are at risk of acute poisoning (neurotoxin inhibiting acetylcholinesterase); chronic effects (memory loss, depression); suspected carcinogens	2
02.3	• Pyrethroids are highly toxic to fish (even though mammal toxicity is low)	1
02.4	<i>Any 3 from:</i> • Neonicotinoids are water soluble so they are absorbed by the roots (from the soil) and translocated through the plant (systemic) • They are moderately persistent so remain in the soil long enough to be absorbed • Seed coating avoids spraying so no spray drift / no dust onto flowers or surrounding habitats • So bees / pollinators are less exposed than when the crop is sprayed	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-265; AQA spec 3.4.3.2.8.	
03.1	• 38.0 / 0.002 • = 19 000 (19 000 times / 1.9×10^4)	2
03.2	• Biomagnification (accept bioaccumulation as part of the process only if biomagnification is also implied by increase up the food chain)	1
03.3	• Persistence: the pesticide is not broken down (by light, microbes, enzymes) so it remains in the environment / organisms long enough to pass along the food chain • Liposolubility: the pesticide dissolves in lipids / fat so it is stored in body tissue and not excreted in (water-based) urine • Each organism eats many organisms from the level below and retains the pesticide so concentration rises at each trophic level • Herons have long lives so accumulate over time	4
03.4	<i>Any 3 from:</i> • Organochlorines killed non-target insects (bees, butterflies) and top predators (birds of prey, otters) through biomagnification • They persist for decades so the damage continued long after use • Use in houses (on walls / nets) does not enter food chains / no runoff into water, so does not biomagnify • Benefit of preventing malaria deaths outweighs the risk (risk:benefit)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-264, 266; AQA spec 3.4.3.2.8.	

Question	Marking guidance	Marks
04.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Neonicotinoids are neurotoxins similar to nicotine; they inhibit the neurotransmitter acetylcholine • High toxicity to insects but low toxicity to vertebrates because they cannot cross the vertebrate blood-brain barrier • Very toxic to bees: sub-lethal doses reduce navigation ability, foraging and immunity to disease; linked to colony decline • Water soluble and moderately persistent: remain in soil, absorbed by roots and translocated (systemic), so present in pollen and nectar • Persistence may allow build-up in aquifers / soil; broken down fastest in sunlight and by soil microbes • Synergism with some fungicides increases toxicity to bees • Exposure greatest when sprayed onto crops or when seed-coating dust is blown by wind • Control: seed coating rather than spraying; spraying when crops are not in flower / at night; still days to reduce drift • Restrictions on use (EU bans on outdoor use on flowering crops); use of alternative pest control (cultural, biological, integrated control) • Evaluation: reduced pesticide use but neonicotinoid alternatives (pyrethroids) also harm pollinators; trade-off with crop yields	9
04.2	<i>Any 1 from:</i> • Biological control (introduced predators / pathogens) • Predator habitats (hedgerows, beetle banks) • Crop rotation • Companion / barrier crops • Sterile male technique / pheromone traps • Genetic resistance	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 265-266; AQA spec 3.4.3.2.8.	
05.1	• The dose that kills 50% of the (test) population	1
05.2	• Insecticide X: LD50 for the beetle (6) is very similar to that of the pest (5) so it has low specificity / kills the predator as well • Loss of the predator removes natural control so the pest population may recover faster / increase (resurgence) • Y has a much higher LD50 for the beetle (150) so few predators killed / more specific	3

Question	Marking guidance	Marks
05.3	<i>Any 2 from:</i> • High water solubility so it is leached into rivers / groundwater / runoff • Toxic to aquatic organisms downstream • Mobile so it can affect areas away from the field • Washed off the crop quickly so more frequent application needed	2
05.4	• Long half-life = high persistence so it is not broken down quickly • So it remains in the organism / environment long enough to build up in tissues (especially if liposoluble)	2
05.5	<i>Any 2 from:</i> • Absorbed and translocated within the plant so protects new growth • Not washed off by rain • Does not need to cover all leaf surfaces / less spray needed • Less spray drift so less harm to non-target species	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-264; AQA spec 3.4.3.2.8.	
06.1	• The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically)	1
06.2	<i>Any 4 from:</i> • Light: UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions • Temperature: higher temperatures increase the rate of chemical breakdown / microbial decomposition • Soil microbes / oxygen: aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils • pH / presence of other chemicals may increase or decrease the rate of degradation • Adsorption onto clay / organic matter may protect from degradation and increase persistence	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
07.1	• Numbers fell from about 210 pairs in 1945 to a minimum of about 25 in 1965 (steepest fall 1955-1960) • Numbers then recovered / rose steadily after 1970, reaching about 190 by 1985 (not yet back to 1945 level)	2
07.2	• $(200 - 25) / 200 \times 100$ • = 87.5%	2
07.3	<i>Any 4 from:</i> • DDT is persistent (remains for decades) and liposoluble so it bioaccumulates in prey (small birds) and biomagnifies in sparrowhawks (top predator) • Sub-lethal doses caused thin eggshells / reduced breeding success so the population declined even without adult deaths • Higher doses caused sterility / death • After restrictions, environmental concentrations slowly fell (persistence explains the delay before recovery) • Breeding success recovered so the population rose	4

Question	Marking guidance	Marks
07.4	<i>Any 2 from:</i> • DDT persists in soil / sediments / food chains for many years so exposure continued • Birds of prey are k-selected: few young per year, slow to mature so populations grow slowly • Other pressures (habitat loss, persecution)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	
08.1	• B - Low liposolubility and low persistence	1
08.2	• Bioaccumulation: build-up of a pollutant in the tissues of an individual organism over time because uptake is faster than excretion / breakdown • Biomagnification: increase in concentration of the pollutant at successive trophic levels of a food chain	2
08.3	• They are intended to be toxic (to the target species) • They are deliberately released / spread into the environment (rather than being an accidental or waste release)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	